



Smart Train Seating & Passenger Comfort System

Improving commuter experience using sensors and automation.

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Task: 9.1P Project.



Problem in Focus

Overcrowding

Trains get packed during peak hours causing discomfort.

Seating Scarcity

Passengers face difficulty finding available seats.

Uneven Distribution

Unequal crowd spread across coaches leads to delays.

Safety Risks

Overcrowding causes boarding delays and potential hazards.

Proposed Solution

Sensor Integration

Pressure and motion sensors monitor seat and standing areas.

Real-time Monitoring

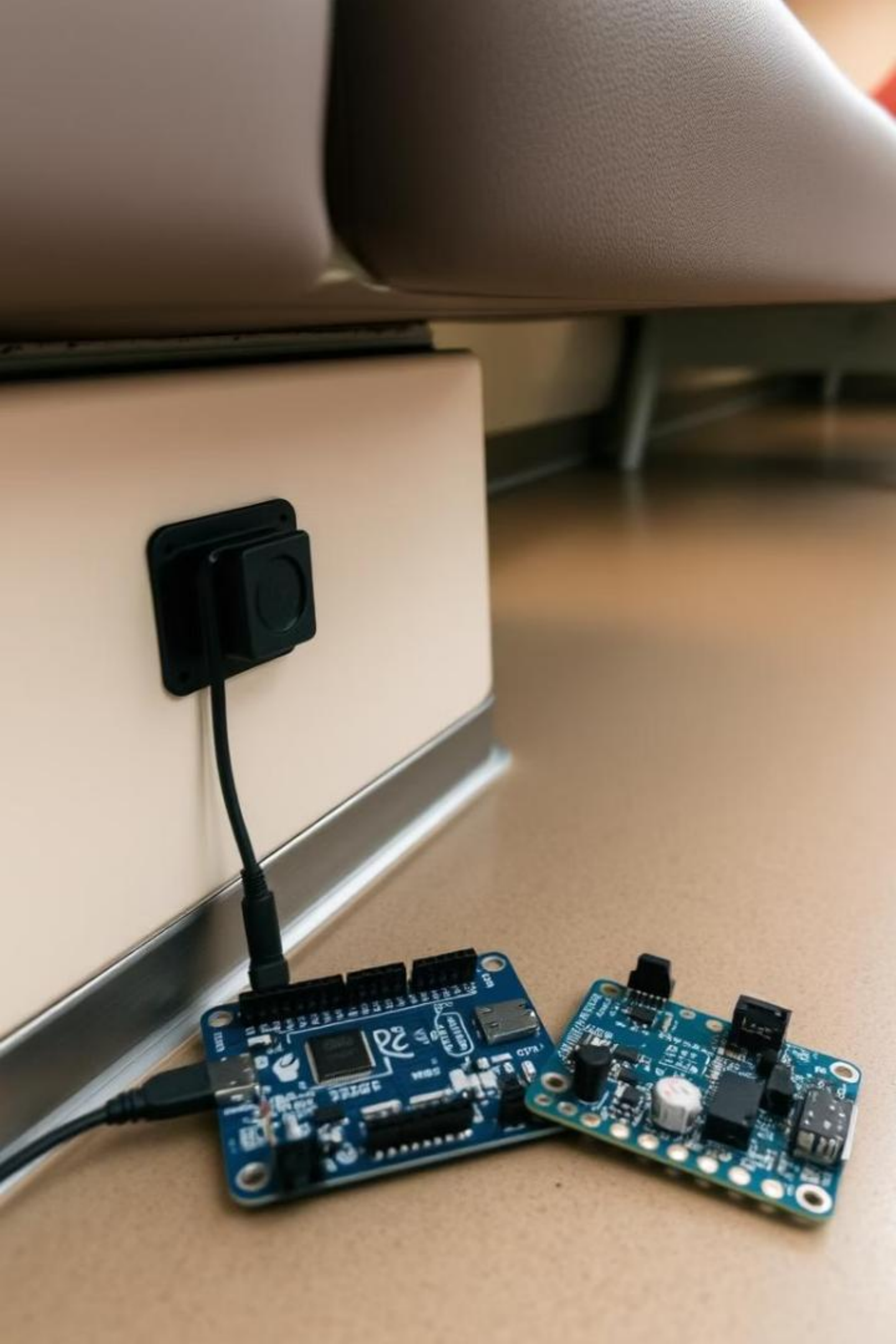
Data shown on coach displays and user mobile apps instantly.

Automated Guidance

Audio announcements direct passengers to less crowded spots.

Privacy & Cost

Low-cost system avoids cameras, ensuring passenger privacy.



How It Works

1

Data Capture

Sensors detect seat occupancy and crowd density.

2

Processing

ESP32 microcontroller processes sensor input rapidly.

3

Data Transmission

Updates sent wirelessly via Wi-Fi to system displays and app.

4

Passenger Alerts

Audio messages guide passengers to balanced areas.

System Architecture



Sensors

Pressure and motion units monitor coaches



Microcontroller

ESP32 processes sensor data



Cloud

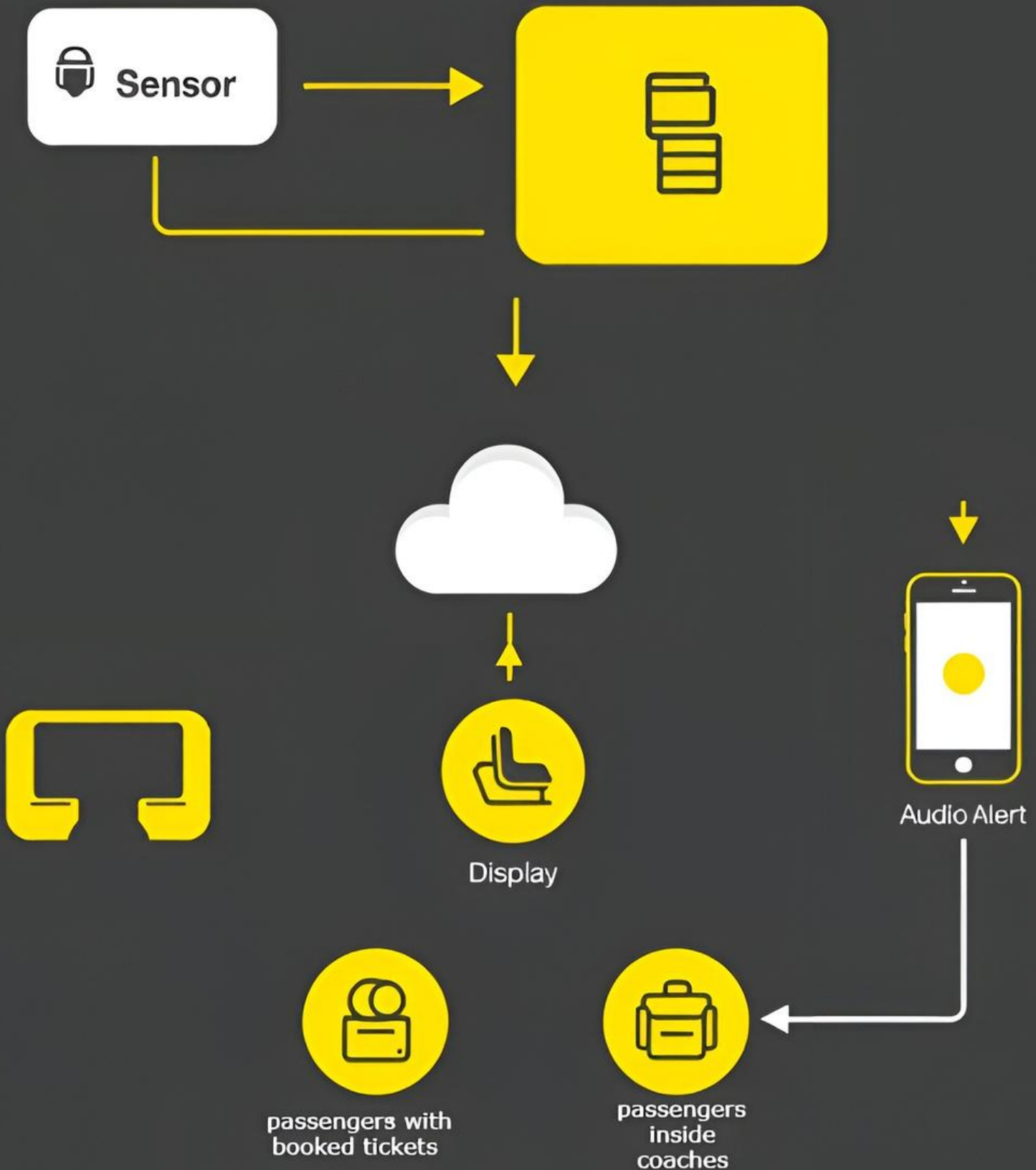
Data synced and stored in real-time



Outputs

Coach displays, mobile apps, audio alerts

Smart train system



Core Technologies Used



ESP32
Microcontro
ller

High efficiency
processing
with Wi-Fi
capability



Pressure
Sensors

Detect seat
occupation
accurately



PIR Motion
Sensors

Track standing
passenger
density



Mobile
Interface

Displays real-
time
occupancy
details to users



Benefits of the System



Efficient Boarding

Reduces delay with real-time seating info



Space Optimization

Distributes passengers evenly across coaches



Safety Enhancement

Limits overcrowding, improving passenger comfort



Privacy Protected

No cameras, fully anonymous occupancy detection



Challenges & Considerations

Sensor Calibration

Must adjust for train motion and vibration noise.

1

2

Signal Reliability

Wi-Fi disruptions may delay real-time updates.

4

3

Installation

Hardware must endure harsh train conditions.

User Behavior

Some passengers might ignore directional audio alerts.

Future Upgrades

Predictive Analytics

Forecast crowding trends for proactive management.

AI Seating Suggestions

Smart recommendations optimizing passenger comfort.



App Alerts

Notify users before boarding crowded trains.

Multi-language Support

Inclusive announcements for diverse passengers.

Solar Sensors

Eco-friendly, self-powered sensor units.

Project Summary & Reflection

Innovative Design

Combines sensors and automation for real-time crowd control.

Ethical Focus

Prioritizes privacy by avoiding camera use.

Scalable Solution

Adaptable to other public transportation modes.

Student Impact

Real-world application of data capture technologies.